

Tata Play Fiber Self-Onboarding Middleware Backend

Solution Design Document (SDD)

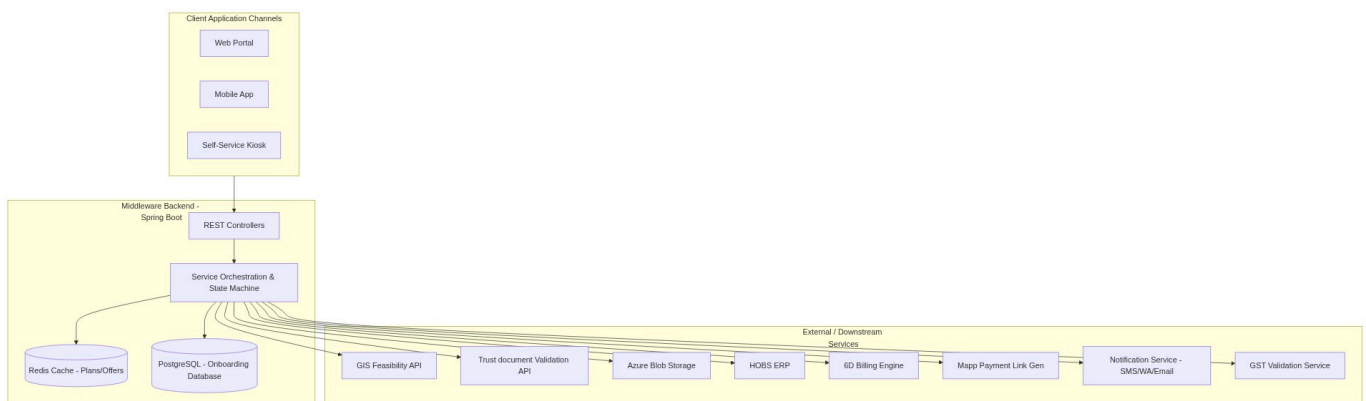
This document describes the high-level solution design and business workflow for the Tata Play Fiber Self-Onboarding Middleware Backend. The middleware acts as a centralized service orchestrating the onboarding journey for customers across third-party client applications.

For low-level technical specifications, database schemas, API JSON payload contracts, and Spring Boot component codes, refer to the Low-Level Design (LLD).

1. High-Level System Architecture

The middleware exposes secure endpoints for frontend applications (web portals, mobile apps, kiosks) and integrates with core internal and external services.

--- Flow Diagram 1 ---



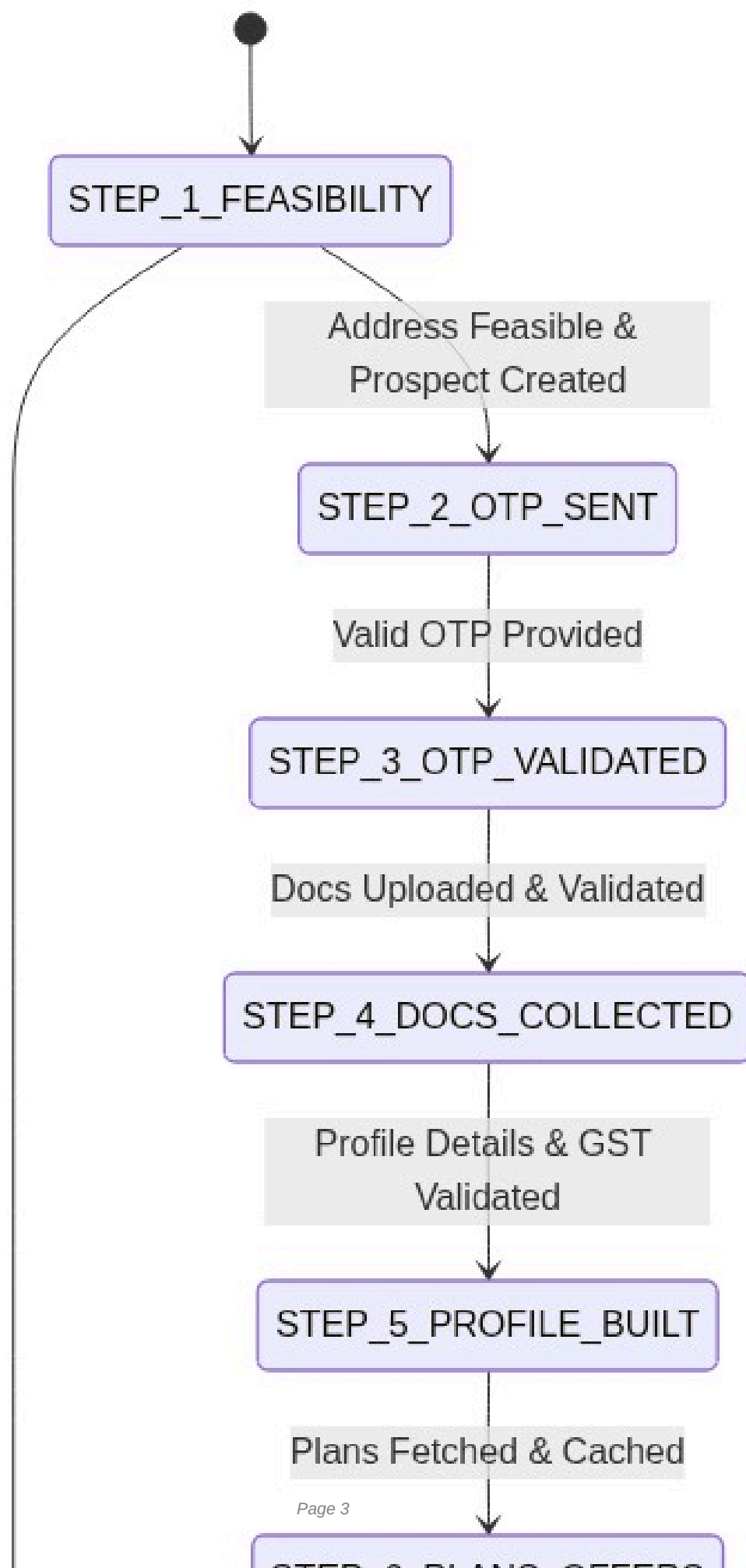
--- Onboarding Flow Table 1 ---

No.	Trigger / Component	Target / Integration	Action Description	Execution Type	Exception Handling
1	3rd Party UI	Middleware API	POST /payment-link	Synchronous	Standard validation error return.
2	Middleware API	HOBS ERP	Create Customer & Billing Sync	Synchronous	Halt & fail request if HOBS is offl..
3	Middleware API	PostgreSQL DB	Update master table to HOBSCREATED	ASYNCHRONOUS	Saves database state.
4	Middleware API	3rd Party UI	Return HTTP 202 Accepted & Account ..	Synchronous	Provides immediate response.
5	Middleware API	6D Billing Engine	Poll billing sync status (3 retries)	Asynchronous	Log sync fail and trigger email ale..
6	Middleware API	Mapp API	POST /payment/link (Generate payment..	Asynchronous	Lock progress and log failure.
7	Middleware API	PostgreSQL DB	Save payment link, amount & sync status	Asynchronous	Updates payment record.
8	Middleware API	Notification Service	Trigger SMS, WA, Email with payment link	Asynchronous	Dispatches link to customer.

2. Onboarding Workflow & State Machine

The onboarding journey is governed by a strict step-by-step flow. Customers must complete each step successfully before being allowed to proceed to the next stage. If they leave the portal and return later, they are automatically restored to their last successful step.

--- Flow Diagram 2 ---



--- Onboarding Flow Table 2 ---

Step	Onboarding Gate / Phase	Key Business Process & System	Next Transition Criteria
Step 1	Feasibility Check	Capture personal details and call G..	If Feasible: Create Prospect ID and..
Step 2	OTP Dispatch	Generate a 6-digit OTP code and sen..	OTP sent successfully -> Proceed to..
Step 3	OTP Validation & Recovery	Check OTP and determine if customer..	If New: Redirect to Step 4. If Retu..
Step 4	Document Collection	Upload POA/POI files to Azure Blob ..	If Trust validation passes -> Proce..
Step 5	Profile Building	Capture additional profile properti..	GST validated or skipped -> Proceed..
Step 6	Plans & Offers Consolidation	Fetch packages based on city (Redis..	Plans consolidated -> Proceed to St..
Step 7	Plan Selection	User selects a package. This config..	Plan selected -> Proceed to Step 8..
Step 8	Amount Calculation	Generate final invoice amount with ..	Invoice generated -> Proceed to Ste..
Step 9	Payment & Account Orchestration	Create account in HOBS (Sync). Wait..	Payment link generated -> Dispatched..
Step 10	Document Migration	Triggered by payment success. Copy..	Files copied successfully -> Procee..
Step 11	WhatsApp Consent	Dispatch interactive consent reques..	Consent approved -> Proceed to Step..
Step 12	CAF PDF Generation	Compile customer details, selected ..	CAF generated -> Proceed to Step 13..
Step 13	EKYC Process Initiation	Create a service ticket in HOBS sys..	Ticket created -> Complete onboardi..

3. High-Level Step Descriptions

Step 1: Capture Lead & GIS Feasibility Check

- **Business Goal**: Validate if the customer's physical location is feasible for fiber provisioning.
- **Process**: Capture name, contact details, email, and coordinates. Call the GIS Feasibility API.
- **If Feasible**: A unique `prospect\_id` is created, lead details are stored as active, and the flow moves to Step 2.
- **If Not Feasible**: Lead details are stored with status `NRFS` (Not Ready For Service) to allow future follow-ups, and the client application is notified to display a feasibility error.

Step 2: Send OTP

- **Business Goal**: Verify customer contact information before collecting sensitive documents.
- **Process**: Generate a secure verification code and send it to the customer's registered mobile number (RMN) and email address.

Step 3: Validate OTP & State Recovery

- **Business Goal**: Authenticate the customer and retrieve their previous session state.
- **Process**: Check the user-entered OTP.
- **New Customers**: Directed to start at Step 4 (Document Collection).
- **Returning Customers**: The middleware checks the master tracking record and resumes their session, returning the data and redirecting them to the exact step where they previously stopped (e.g., Plan Selection or Payment).

Step 4: Document Collection & Validation

- **Business Goal**: Gather identification and address proof, and verify authenticity.
- **Process**:
 - * Upload Proof of Identity (POI), Proof of Address (POA), and a live photo to Azure Blob Storage (retaining file paths and metadata only in the database).
 - * A declaration form is auto-generated in PDF format and saved with the files.
 - * Call verification APIs to perform OCR validation and facial matching. If successful, progress to Step 5 is unlocked.

Step 5: Profile Building

- **Business Goal**: Complete customer billing details and business info.
- **Process**: Capture customer demographic and GST details. Validate GST registrations against national portals.

Step 6: Plans & Offers Consolidation

- **Business Goal**: Present available fiber packages and active offers.
- **Process**: Retrieve packages based on the customer's city (cached in Redis to ensure low latency) and merge them with active promotional coupons/offers fetched from the Magix API.

Step 7: Plan Selection (Overridable)

- **Business Goal**: Record the customer's selected fiber plan.
- **Process**: Persist package and discount choices. The customer can go back and re-select or override their selection at any time before payment.

Step 8: Calculate Amount

- **Business Goal**: Display a clear pricing breakdown to the customer.
- **Process**: Compute base prices, apply promotional discounts, calculate local taxes (GST), and return the final invoice amount.

Step 9: Payment Link Generation & Provisioning Orchestration

- **Business Goal**: Create customer accounts in core systems and provide a payment route.
- **Process**:
 - * Synchronously create the customer account and billing account inside the HOBS ERP.
 - * Asynchronously poll the 6D Billing Engine to verify synchronization.
 - * Once synced, invoke the Mapp API to generate a single-use payment link and send it via SMS, email, and WhatsApp.
 - * **Regeneration Logic**: If the link expires or the request is retried, the middleware checks HOBS/6D creation. If not created, it retries provisioning; otherwise, it sends the existing unexpired link or generates a fresh one.

Step 10: Document Migration

- **Business Goal**: Transition documents from lead storage to permanent customer storage.

- **Process**: Upon receiving a payment success notification (via webhook), documents are automatically copied from the prospect folder to the permanent customer ID folder in Azure Blob Storage.

Step 11: Customer WhatsApp Consent

- **Business Goal**: Gather statutory electronic consent.
- **Process**: Send a template message to the customer's WhatsApp. The customer grants consent directly via WhatsApp interactive buttons, which triggers a webhook to update the onboarding record.

Step 12: CAF Generation

- **Business Goal**: Finalize the Customer Application Form.
- **Process**: Generate the final CAF document as a PDF containing all validated details, photo, and consent logs, and upload it to the customer ID folder.

Step 13: EKYC Start

- **Business Goal**: Initiate the final validation ticket.
- **Process**: Create a service ticket in HOBS to notify field/logical verification teams to start the EKYC process, concluding the onboarding cycle.

4. Key Business Controls

State Gatekeeper (Sequential Verification)

A core component ensures customers cannot jump steps or access endpoints ahead of their current authorized stage. Each request verifies the prospect's stage in the database. Out-of-order calls are immediately rejected, returning instructions to return to the active stage.

Document Access Security

No document files are accessible via public links. The middleware serves documents via temporary, secure Shared Access Signature (SAS) tokens with a 10-minute expiry time to prevent unauthorized access.

Status Tracking & Retrieval API

A unified endpoint retrieves the current milestone and status details using either the prospect ID, customer ID, or registered mobile number (RMN).